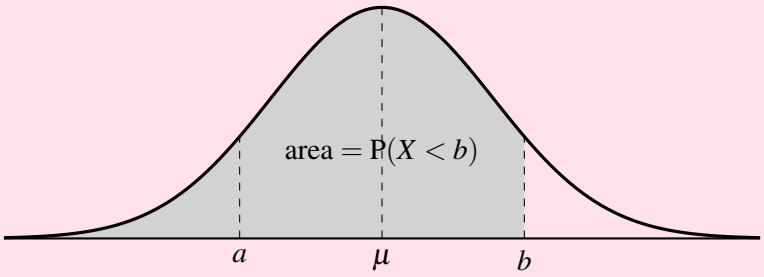
Normal Distribution

Inverse Normal & Finding μ and σ

IB Maths AA SL · Notes · Questions · Worked Answers

1. Finding a boundary value
2. Finding σ or μ (one unknown)
3. Finding both μ and σ
4. Conditional probability, expected numbers & binomial



No calculator Calculator (TI-Nspire CX II) allowed

Every item in the Index is a clickable link. Each topic has its own colour.

Index

= no calculator = calculator allowed

Sources: Exam questions: IB Mathematics SL / AA SL Paper 2 (session · paper/time zone · question), most recent first; **Worksheet** = AA SL *Finding the Mean and Standard Deviation for Normal Distributions*; **RV** = Revision Village (Practice Set 2 numbering); **SME** = Save My Exams IB *Normal Distribution*; **Oxford** = Oxford IB textbook Ex 14L, 14M and exam-style questions; **Kognity** = Kognity practice sets; **Y13 Worksheet** = IB Year 13 Binomial & Normal worksheet; **School HW** = school homework extension problems; **Practice** = written for this booklet. Repeated questions appear only once (several homework questions are already in the *Normal Distribution* booklet).

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Questions

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4. Conditional Probability, Expected Numbers & Binomial

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Key Notes

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1. The normal distribution — a CONTINUOUS distribution

$X \sim \mathrm{N}(\mu, \sigma^2)$: bell-shaped, symmetric about the mean μ; mean = median = mode; total area $= 1$.

Because X is **continuous**, the probability of one exact value is 0, so

$$\mathrm{P}(X > 80) = \mathrm{P}(X \geq 80) \qquad \mathrm{P}(80 < X < 86) = \mathrm{P}(80 \leq X \leq 86).$$

★ **Use the boundary value exactly as given.** For "between 80 and 86" the lower bound is **80** (not 81); for "more than 80" the lower bound is still **80**: `normCdf(80, 9E99, μ, σ)`.
Only for a **discrete** (binomial) variable does "more than 80" become $X \geq 81$.

2. Inverse normal — always use the LEFT-tail area

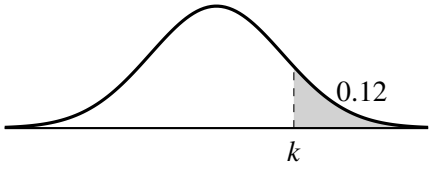
Inverse Normal on the TI-Nspire CX II gives the value k with a given area **to its left**: $\mathrm{P}(X < k) =$ area.

★ **If the area given is a right tail**, change it first:

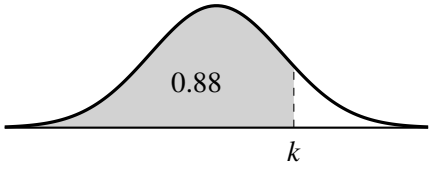
$\mathrm{P}(X > k) = 0.12 \Rightarrow \mathrm{P}(X < k) = 1 - 0.12 = 0.88$

then `invNorm(0.88, μ, σ)`.

"5% are taller than 1.70 m" ⇒ use 0.95.
"Between μ and k is 35%" ⇒ use $0.5 + 0.35 = 0.85$.



given: right tail



use: left tail

3. Standardising: the z-score

$$z = \frac{x-\mu}{\sigma} \qquad \Longleftrightarrow \qquad x = \mu + z\sigma, \qquad Z \sim \mathrm{N}(0,1)$$

One unknown (σ or μ): find $z = \text{invNorm(left area, 0, 1)}$, then substitute into $z = \dfrac{x-\mu}{\sigma}$ and solve.

e.g. $\mu = 16$, $\mathrm{P}(X < 20) = 0.85$: $z = 1.03643$, $1.03643 = \dfrac{20-16}{\sigma} \Rightarrow \sigma = 3.86$.

(You cannot use invNorm with μ and σ when one of them is unknown — use z.)

4. Both μ and σ unknown — simultaneous equations

1. Write both facts as **left-tail** probabilities: $\mathrm{P}(X < x_1) = p_1$, $\mathrm{P}(X < x_2) = p_2$.
2. $z_1 = \text{invNorm}(p_1, 0, 1)$, $z_2 = \text{invNorm}(p_2, 0, 1)$ (keep 6 s.f.).
3. $\mu + z_1\sigma = x_1$ and $\mu + z_2\sigma = x_2$; solve (menu → 3: Algebra → 2: Solve System of Linear Equations).

Check: $\sigma > 0$, and μ lies between the two values when one z is negative and one positive.

5. Conditional probability with the normal distribution

$$\mathrm{P}(A \,|\, B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \qquad \text{e.g.} \quad \mathrm{P}(T < 82 \,|\, T > 80) = \frac{\mathrm{P}(80 < T < 82)}{\mathrm{P}(T > 80)}$$

Draw the curve, shade the "given" region B, then the part of it that is also in A. Independent events: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$.

6. Expected numbers and the link to the binomial

Expected number $= n \times p$, where p comes from the normal distribution.

★ **Do NOT round an expected value to a whole number.** Even when it counts flights, apples or students, $\mathrm{E}(X)$ is an average: $64 \times 0.0711930 = \mathbf{4.56}$ flights (not 5). Round only when the question asks for a whole number.

"How many **out of** n" with a probability question ⇒ binomial: $X \sim \mathrm{B}(n, p)$ with p from the normal.
Then the binomial is discrete: $\mathrm{P}(X > 6) = \mathrm{P}(X \geq 7) = 1 - \mathrm{P}(X \leq 6)$.

7. TI-Nspire CX II key steps

Probability (area)	menu → 5: Probability → 5: Distributions → **2: Normal Cdf** (Lower, Upper, μ, σ; use $\pm$9E99 for $\pm\infty$)
Value from an area	menu → 5 → 5 → **3: Inverse Normal** (**left-tail** area, μ, σ)
z-score	Inverse Normal with $\mu = 0$, $\sigma = 1$
Binomial	5 → 5 → **D: Binomial Pdf** (=) or **E: Binomial Cdf** (range)
Two equations	menu → 3: Algebra → 2: Solve System of Linear Equations

Store unrounded answers (ctrl var) and give final answers to 3 s.f.

8. Exam checklist

- Sketch the curve, mark μ and the value(s), shade the area you need.
- Write the probability statement before using the GDC, e.g. $\mathrm{P}(X < k) = 0.625$.
- Right tail given? Convert to the left tail. Boundaries are used exactly (continuous).
- Expected values stay as decimals. Final answers to 3 s.f.

1 answer → Calculator **Nov 2013 · P2 · Q6**

[Maximum mark: 7]

The time taken for a student to complete a task is normally distributed with a mean of 20 minutes and a standard deviation of 1.25 minutes.

(a) A student is selected at random. Find the probability that the student completes the task in less than 21.8 minutes. [2]

(b) The probability that a student takes between k and 21.8 minutes is 0.3. Find the value of k. [5]

2 answer → Calculator **Practice**

[Maximum mark: 4]

$X \sim \mathrm{N}(50, 6^2)$. Given that $\mathrm{P}(X > k) = 0.12$, find the value of k.

1 ← question *Nov 2013 · P2 · Q6*

(a) $P(T < 21.8) = \boxed{0.925}$ $(z = 1.44)$ **(M1) A1**

(b) $P(T < 21.8) - P(T < k) = 0.3$ **(M1)**

$P(T < k) = 0.925 - 0.3 = 0.625$ **(A1)**

$k = \text{invNorm}(0.625, 20, 1.25)$, $(z = 0.3186)$ **(M1)(A1)**

$\boxed{k = 20.4}$ **A1**

TI-Nspire CX: Question 1

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) `normCdf(−∞,21.8,20,1.25)` $= 0.925$. Then menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.625,20,1.25)` $= 20.4$.

2 ← question *Practice*

Right-tail area is given, so change it to the **left tail**: $P(X < k) = 1 - 0.12 = 0.88$ **M1 A1**

$k = \text{invNorm}(0.88, 50, 6) = \boxed{57.0}$ $(57.0499\ldots)$ **A1 A1**

Common error: `invNorm(0.12,50,6)` gives 42.95 — the point with 12% to its *left*.

TI-Nspire CX: Question 2

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.88,50,6)` $= 57.05$. The TI-Nspire CX II Inverse Normal uses the **left-tail area** unless you choose a tail setting, so convert first.

3 answer → Calculator **Kognity A · Q2**

[Maximum mark: 2]

In a season of sport fishing, the mean weight of halibut caught is 16 kg and the standard deviation is 2.3 kg. The probability that a randomly caught halibut weighs less than x is 0.16. What is the value of x?

4 answer → Calculator **Kognity A · Q3**

[Maximum mark: 2]

In a season of sport fishing, the mean weight of halibut caught is 16 kg and the standard deviation is 2.3 kg. The probability that a randomly caught halibut weighs **more** than x is 0.975. Correct to 2 significant figures, what is the value of x?

5 answer → Calculator **Kognity A · Q4**

[Maximum mark: 4]

A soft drink bottling company fills bottles with volumes $X \sim \mathrm{N}(2000, 1.5^2)$ millilitres.

(a) The company finds that 1% of the bottles are extremely underfilled. To the nearest tenth of a millilitre, find the value that determines an extremely underfilled bottle. [2]

(b) The company finds that 45.22% of the bottles are filled between 2000 millilitres and b millilitres. Find the value of b rounded to the nearest tenth of a millilitre. [2]

3 ← question *Kognity A · Q2*

Left tail given: $x = \text{invNorm}(0.16, 16, 2.3) = \boxed{13.7}$ kg **M1 A1**

TI-Nspire CX: Question 3

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.16,16,2.3)`

4 ← question *Kognity A · Q3*

Right tail $0.975 \Rightarrow$ left tail $1 - 0.975 = 0.025$ **M1**

$x = \text{invNorm}(0.025, 16, 2.3) = 11.49 \approx \boxed{11}$ kg (2 s.f.) **A1**

TI-Nspire CX: Question 4

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.025,16,2.3)`

5 ← question *Kognity A · Q4*

(a) $\text{invNorm}(0.01, 2000, 1.5) = \boxed{1996.5}$ **M1 A1**

(b) $\text{P}(X < b) = 0.5 + 0.4522 = 0.9522$ **M1** $b = \boxed{2002.5}$ **A1**

TI-Nspire CX: Question 5

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.01,2000,1.5)`; `invNorm(0.9522,2000,1.5)`

6 answer → Calculator **Kognity B · Q2**

[Maximum mark: 2]

The random variable X is normally distributed with mean $\mu = 5.9$ and standard deviation $\sigma = 1.3$. Find the value of k if $P(X < k) = 0.19$. Give your answer to three significant figures.

7 answer → Calculator **Kognity C · Q5**

[Maximum mark: 1]

Let $Z \sim N(0, 1)$. Use your GDC to find c when $P(Z < c) = 0.3$. Give your answer correct to 3 significant figures.

8 answer → Calculator **Y13 Worksheet · Q8**

[Maximum mark: 3]

A factory produces screws with lengths normally distributed, mean 3.00 cm and $\sigma = 0.05$ cm. Find the length above which only 2% of screws lie.

9 answer → Calculator **Y13 Worksheet · Q9**

[Maximum mark: 2]

The heights of students are normally distributed with mean 170 cm and $\sigma = 6$ cm. Find the height corresponding to the 90th percentile.

6 ← question *Kognity B · Q2*

$k = \text{invNorm}(0.19, 5.9, 1.3) = \boxed{4.76}$ **M1 A1**

TI-Nspire CX: Question 6

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.19,5.9,1.3)`

7 ← question *Kognity C · Q5*

$c = \text{invNorm}(0.3, 0, 1) = \boxed{-0.524}$ **A1**

TI-Nspire CX: Question 7

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.3,0,1)`

8 ← question *Y13 Worksheet · Q8*

$\text{P}(X > x) = 0.02 \Rightarrow \text{P}(X < x) = 0.98$ **M1**

$x = \boxed{3.10}\,\text{cm}$ **A2**

TI-Nspire CX: Question 8

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.98,3,0.05)`

9 ← question *Y13 Worksheet · Q9*

$x = \text{invNorm}(0.9, 170, 6) = \boxed{177.7}\,\text{cm}$ **M1 A1**

TI-Nspire CX: Question 9

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.9,170,6)`

10 answer → Calculator **Oxford Ex 14L · Q1**

[Maximum mark: 2]

$X \sim \mathrm{N}(5.5, 0.2^2)$ and $\mathrm{P}(X > a) = 0.235$. Find the value of a.

11 answer → Calculator **Oxford Ex 14L · Q2**

[Maximum mark: 4]

The masses, M, of tins of dog food are such that $M \sim \mathrm{N}(420, 10^2)$. Find:

(a) the first quartile [2]

(b) the 90th percentile. [2]

12 answer → Calculator **Oxford Ex 14L · Q3**

[Maximum mark: 7]

Regulations in a country insist that all mineral bottles that claim to contain 500 ml must have at least that amount. "Yummy Cola" has a machine for filling bottles, which puts a mean of 502 ml into each bottle, with a standard deviation of 1.6 ml. This follows a normal distribution.

(a) An inspector randomly selects a bottle of "Yummy Cola". Find the probability that the selected bottle violates the regulations. [2]

(b) Find the proportion of bottles which contain between 500 ml and 505 ml. [2]

(c) 95% of bottles contain between a ml and b ml of liquid where a and b are symmetrical about the mean. Find the values of a and b. [3]

10 ← question *Oxford Ex 14L · Q1*

$\mathrm{P}(X < a) = 0.765;\ a = \boxed{5.64}$ **M1 A1**

TI-Nspire CX: Question 10

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.765,5.5,0.2)`

11 ← question *Oxford Ex 14L · Q2*

(a) $\mathrm{invNorm}(0.25, 420, 10) = \boxed{413}$ (b) $\mathrm{invNorm}(0.9, 420, 10) = \boxed{433}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 11

`invNorm(0.25,420,10)`, `invNorm(0.9,420,10)`

12 ← question *Oxford Ex 14L · Q3*

(a) $\mathrm{P}(X < 500) = \boxed{0.106}$ **M1 A1**

(b) $\boxed{0.864}$ (86.4%) **M1 A1**

(c) 2.5% each tail: $a = \boxed{498.9}$, $b = \boxed{505.1}$ **M1 A1 A1**

TI-Nspire CX: Question 12

`invNorm(0.025,502,1.6)`, `invNorm(0.975,502,1.6)`

13 answer → Calculator Oxford Ex 14L · Q4

[Maximum mark: 4]

The masses of heads of lettuce sold at a hypermarket are normally distributed with a mean mass of 550 g and standard deviation of 25 g.

(a) If a head of lettuce is chosen at random, find the probability that its mass lies between 520 g and 570 g. [2]

(b) Find the mass exceeded by 10% of the heads of lettuce. [2]

14 answer → Calculator Oxford Ex 14L · Q5

[Maximum mark: 4]

The marks of 500 candidates in an examination are normally distributed with a mean of 55 marks and a standard deviation of 15 marks.

(a) If 5% of the candidates obtain a distinction by scoring d marks or more, find the value of d. [2]

(b) If 10% of students fail by scoring f marks or less, find the value of f. [2]

15 answer → Calculator SME Medium · Q4

[Maximum mark: 8]

(a) The random variable $X \sim \mathrm{N}(330, 10^2)$. Find the value of a, to 2 decimal places, such that: (i) $\mathrm{P}(X < a) = 0.25$ (ii) $\mathrm{P}(X > a) = 0.25$ (iii) $\mathrm{P}(315 \leq X \leq a) = 0.5$ [4]

(b) The random variable $Y \sim \mathrm{N}(10, 10)$. Find the value of b and the value of c, each to 2 decimal places, such that: (i) $\mathrm{P}(Y < b) = 0.4$ (ii) $\mathrm{P}(Y > c) = 0.25$ [2]

(c) Use a sketch of the distribution of Y to explain why $\mathrm{P}(b \leq Y \leq c) = 0.35$. [2]

13 ← question *Oxford Ex 14L · Q4*

(a) $\boxed{0.673}$ **M1 A1**

(b) left tail 0.9: $\boxed{582}$ g **M1 A1**

TI-Nspire CX: Question 13

```
normCdf(520,570,550,25); invNorm(0.9,550,25)
```

14 ← question *Oxford Ex 14L · Q5*

(a) left tail 0.95: $d = \boxed{79.7}$ **M1 A1**

(b) $f = \boxed{35.8}$ **M1 A1**

TI-Nspire CX: Question 14

```
invNorm(0.95,55,15); invNorm(0.1,55,15)
```

15 ← question *SME Medium · Q4*

(a)(i) $\boxed{323.26}$ (ii) left tail 0.75: $\boxed{336.74}$ **A1 A1**

(iii) $\mathrm{P}(X < a) = \mathrm{P}(X < 315) + 0.5 = 0.06681 + 0.5 = 0.56681$: $a = \boxed{331.68}$ **M1 A1**

(b) $\sigma = \sqrt{10}$ (variance 10): (i) $b = \boxed{9.20}$ (ii) $c = \boxed{12.13}$ **A1 A1**

(c) area left of c is 0.75, left of b is 0.4: $0.75 - 0.4 = 0.35$ **M1 R1**

TI-Nspire CX: Question 15

```
invNorm(0.56681,330,10); invNorm(0.4,10,√(10))
```

16 answer → Calculator **SME Medium · Q5**

[Maximum mark: 5]

(a) The test scores, X, of a group of RAF recruits in an aptitude test are modelled as $X \sim \mathrm{N}(210, 27.8^2)$. (i) Find the values of a and b such that $\mathrm{P}(X < a) = 0.25$ and $\mathrm{P}(X > b) = 0.25$. (ii) Hence find the interquartile range of the scores. [3]

(b) Those who score in the top 30% on the test move on to the next stage of training. One of the recruits, Amelia, achieves a score of 231. Determine whether Amelia will move on to the next stage of training. [2]

16 ← question *SME Medium · Q5*

(a)(i) $a = \boxed{191.2}$, $b = \boxed{228.8}$ (ii) IQR $= \boxed{37.5}$ **A1 A1 A1**

(b) cut-off $=$ invNorm$(0.7, 210, 27.8) = 224.6$; $231 > 224.6$: **yes** **M1**

R1

TI-Nspire CX: Question 16

```
invNorm(0.25,210,27.8); invNorm(0.7,210,27.8)
```

17 answer → No calculator **SME Medium · Q7**

[Maximum mark: 5]

The table shows the percentage points of the normal distribution. The values z in the table are those which a random variable $Z \sim \mathrm{N}(0,1)$ exceeds with probability p.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0365	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

(a) (i) Use the table to find the value of z for which $\mathrm{P}(Z > z) = 0.2$. (ii) Hence work out the values of a and b for which $\mathrm{P}(Z < a) = 0.2$ and $\mathrm{P}(Z < b) = 0.8$. [3]

(b) The weights, W kg, of coconuts are modelled as a normal distribution with mean 1.25 kg and standard deviation 0.38 kg. The plantation only considers coconuts exportable if their weight falls into the 20% to 80% interpercentile range. Use part (a)(ii) to find the range of possible weights, to the nearest 0.01 kg, for an exportable coconut. [2]

17 ← question *SME Medium · Q7*

(a)(i) $\boxed{0.8416}$ (ii) by symmetry $a = \boxed{-0.8416}$, $b = \boxed{0.8416}$ **B1 B1 B1**

(b) $1.25 \pm 0.8416(0.38)$: $\boxed{0.93 \leq W \leq 1.57}$ kg **M1 A1**

18 answer → Calculator **SME Hard · Q3**

[Maximum mark: 6]

(a) If $X \sim \mathrm{N}(24, 16)$, find x given that: (i) $\mathrm{P}(X < x) = 0.7$ (ii) $\mathrm{P}(X > x) = 0.15$ [3]

(b) Find a and b given that $\mathrm{P}(a < X < b) = 0.95$ and a and b are equal distances from the mean. [3]

18 ← question *SME Hard · Q3*

$\sigma = 4$

(a)(i) $\boxed{26.1}$ (ii) left tail 0.85: $\boxed{28.1}$ **A1 M1 A1**

(b) 2.5% each tail: $a = \boxed{16.2}$, $b = \boxed{31.8}$ **M1 A1 A1**

TI-Nspire CX: Question 18

`invNorm(0.7,24,4)`, `invNorm(0.85,24,4)`, `invNorm(0.025,24,4)`

19 answer → Calculator **SME Hard · Q7**

[Maximum mark: 8]

(a) A software firm has recorded the distance, D, that their users have scrolled with a mouse over the course of one year. The data is distributed normally with a mean of 18.4 miles and a standard deviation of 5.8 miles. Calculate the interquartile range of the data. [4]

(b) It is known that 8% of the users scroll more than d miles but less than 22 miles. Find the value of d. [4]

19 ← question *SME Hard · Q7*

(a) $Q_3 - Q_1 = 22.31 - 14.49 = \boxed{7.82}$ miles **M1 A1 A1 A1**

(b) $\mathrm{P}(D < d) = \mathrm{P}(D < 22) - 0.08 = 0.73260 - 0.08 = 0.65260$ **M1 A1**

$d = \boxed{20.7}$ **M1 A1**

TI-Nspire CX: Question 19

```
normCdf(-9E99,22,18.4,5.8); invNorm(0.6526,18.4,5.8)
```

20 answer →

Calculator

Nov 2017 · P2 · Q7

[Maximum mark: 7]

The heights of adult males in a country are normally distributed with a mean of 180 cm and a standard deviation of σ cm. 17% of these men are shorter than 168 cm. 80% of them have heights between $(192 - h)$ cm and 192 cm.

Find the value of h.

20 ← question *Nov 2017 · P2 · Q7*

$\mathrm{P}(X < 168) = 0.17 \Rightarrow z = \text{invNorm}(0.17) = -0.95416$ **(A1)**

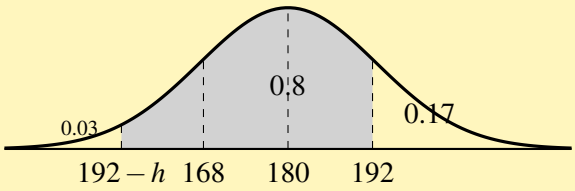
$-0.95416 = \dfrac{168 - 180}{\sigma}$ **(M1)** $\sigma = 12.5765$ **(A1)**

By symmetry $\mathrm{P}(X > 192) = 0.17 \Rightarrow \mathrm{P}(X < 192) = 0.83$ **(A1)**

$\mathrm{P}(X < 192 - h) = 0.83 - 0.8 = 0.03$ **(A1)**

$192 - h = \text{invNorm}(0.03, 180, 12.5765) = 156.346$ **(A1)**

$\boxed{h = 35.7}$ **A1**



TI-Nspire CX: Question 20

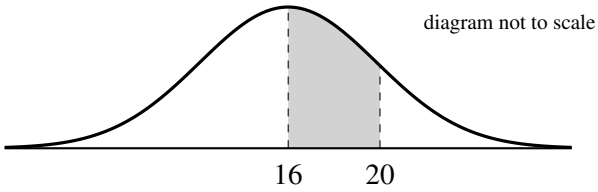
menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. (i) `invNorm(0.17,0,1)` → -0.95416, then $\sigma = -12/-0.95416$. (ii) `invNorm(0.03,180,12.5765)` → 156.346.

21 answer →

Calculator **Worksheet · Q1**

[Maximum mark: 7]

Let the random variable X be normally distributed with mean 16, as shown in the following diagram.



The shaded region between 16 and 20 represents 35% of the distribution.

(a) Find $P(X > 20)$. [2]

(b) Find the standard deviation of X. [5]

22 answer →

Calculator **Worksheet · Q6**

[Maximum mark: 6]

The lengths L, in millimetres, of caterpillars in an enclosure are normally distributed with a mean of 12.5 mm and a standard deviation of σ mm.

(a) Write down the probability that a randomly selected caterpillar has a length greater than 12.5 mm. [1]

(b) It is given that 6.5% of caterpillars have a length greater than 14 mm.

(i) Find the value of σ;

(ii) Find the probability that a randomly selected caterpillar has a length less than 11.5 mm. [5]

21 ← question *Worksheet · Q1*

(a) $\mathrm{P}(X > 16) = 0.5$, so $\mathrm{P}(X > 20) = 0.5 - 0.35 = \boxed{0.15}$ **M1 A1**

(b) $\mathrm{P}(X < 20) = 0.85$ (use the **left** tail) **(M1)**

$z = \text{invNorm}(0.85) = 1.03643$ **(A1)**

$1.03643 = \dfrac{20-16}{\sigma}$ **(M1)(A1)**

$\boxed{\sigma = 3.86}$ $(3.85939\ldots)$ **A1**

TI-Nspire CX: Question 21

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.85,0,1)` $= 1.03643$; then $\sigma = 4 \div 1.03643$.

22 ← question *Worksheet · Q6*

(a) $\boxed{0.5}$ (the mean is in the middle) **A1**

(b)(i) $\mathrm{P}(L > 14) = 0.065 \Rightarrow \mathrm{P}(L < 14) = 0.935$ (left tail) **(M1)**

$z = \text{invNorm}(0.935) = 1.51410$; $1.51410 = \dfrac{14-12.5}{\sigma}$ **(A1)(M1)**

$\boxed{\sigma = 0.991}$ $(0.990686\ldots)$ **A1**

(ii) $\mathrm{P}(L < 11.5) = \text{normCdf}(-\infty, 11.5, 12.5, 0.990686) = \boxed{0.156}$ **A1**

TI-Nspire CX: Question 22

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.935,0,1)` $= 1.51410$. menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) Keep σ unrounded (store it: ctrl var).

23

answer →

Calculator **Practice**

[Maximum mark: 4]

The lifetimes of a type of battery are normally distributed with mean μ hours and standard deviation 15 hours. 10% of batteries last longer than 200 hours.

Find the value of μ.

24

answer →

No calculator **Kognity C · Q1**

[Maximum mark: 3]

Let $X \sim \mathrm{N}(23, 7^2)$ and $Y \sim \mathrm{N}(19, 6^2)$. Given that for some value c the equation $\mathrm{P}(X > c) = \mathrm{P}(Y > 22)$ holds, find the exact value of c. Give your answer as a fraction in the form $\frac{a}{b}$ simplified to its lowest terms.

25

answer →

Calculator **Oxford Ex 14M · Q1**

[Maximum mark: 3]

$X \sim \mathrm{N}(30, \sigma^2)$ and $\mathrm{P}(X > 40) = 0.115$. Find the value of σ.

26

answer →

Calculator **Oxford Ex 14M · Q2**

[Maximum mark: 3]

$X \sim \mathrm{N}(\mu, 4^2)$ and $\mathrm{P}(X < 20.5) = 0.9$. Find the value of μ.

23 ← question *Practice*

$\mathrm{P}(X > 200) = 0.1 \Rightarrow \mathrm{P}(X < 200) = 0.9$ **M1**

$z = \text{invNorm}(0.9) = 1.28155$ **A1**

$1.28155 = \dfrac{200-\mu}{15}$ **M1**

$\mu = 200 - 15(1.28155) = \boxed{181}$ (180.777) **A1**

TI-Nspire CX: Question 23

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.9,0,1)` $= 1.28155$.

24 ← question *Kognity C · Q1*

Same tail $\Rightarrow$ same z: $\dfrac{22-19}{6} = \dfrac{1}{2}$ **M1**

$\dfrac{c-23}{7} = \dfrac{1}{2} \Rightarrow c = 23 + \dfrac{7}{2} = \boxed{\dfrac{53}{2}}$ **M1 A1**

25 ← question *Oxford Ex 14M · Q1*

$z = \text{invNorm}(0.885) = 1.2004;\ \ \sigma = \dfrac{10}{1.2004} = \boxed{8.33}$ **M1 A1 A1**

TI-Nspire CX: Question 25

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.885,0,1)`

26 ← question *Oxford Ex 14M · Q2*

$z = 1.2816;\ \ \mu = 20.5 - 4(1.2816) = \boxed{15.4}$ **M1 A1 A1**

TI-Nspire CX: Question 26

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.9,0,1)`

27 answer → Calculator Oxford Ex 14M · Q5

[Maximum mark: 3]

The mean height of children of a certain age is 136 cm. 12% of children have a height of 145 cm or more. Find the standard deviation of the heights.

28 answer → Calculator Oxford Ex 14M · Q6

[Maximum mark: 3]

The standard deviation of masses of loaves of bread is 20 g. Only 1% of loaves have mass less than 500 g. Find the mean mass of the loaves.

29 answer → Calculator Oxford Ex 14M · Q7

[Maximum mark: 5]

The masses of cauliflowers are normally distributed with mean 0.85 kg. 74% of cauliflowers have mass less than 1.1 kg. Find:

(a) the standard deviation of masses of cauliflowers [3]

(b) the percentage of cauliflowers with mass greater than 1 kg. [2]

30 answer → Calculator Oxford Ex 14M · Q8

[Maximum mark: 3]

The lengths of some boxes are normally distributed with mean μ and standard deviation 7 cm. If 2.5% of the boxes measure more than 68 cm, find the value of μ.

27 ← question *Oxford Ex 14M · Q5*

$z = \text{invNorm}(0.88) = 1.1750;\ \sigma = \dfrac{9}{1.1750} = \boxed{7.66}\,\text{cm}$ **M1 A1 A1**

TI-Nspire CX: Question 27

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.88,0,1)`

28 ← question *Oxford Ex 14M · Q6*

$z = -2.3263;\ \mu = 500 + 20(2.3263) = \boxed{546.5}\,\text{g}$ **M1 A1 A1**

TI-Nspire CX: Question 28

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.01,0,1)`

29 ← question *Oxford Ex 14M · Q7*

(a) $z = 0.6433;\ \sigma = \dfrac{0.25}{0.6433} = \boxed{0.389}\,\text{kg}$ **M1 A1 A1**

(b) $\text{P}(X > 1) = 0.350 \Rightarrow \boxed{35.0\%}$ **M1 A1**

TI-Nspire CX: Question 29

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.74,0,1)`

30 ← question *Oxford Ex 14M · Q8*

$z = \text{invNorm}(0.975) = 1.96;\ \mu = 68 - 7(1.96) = \boxed{54.3}\,\text{cm}$ **M1 A1 A1**

TI-Nspire CX: Question 30

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.975,0,1)`

31 answer → Calculator Oxford Ex 14M · Q9

[Maximum mark: 3]

A roll of wrapping paper is sold as "3 m long". It is found that actually only 35% of rolls are over 3 m long and that the average length of the rolls of wrapping paper is 2.9 m. Find the value of the standard deviation of the lengths of rolls of wrapping paper, assuming that the lengths of rolls follow a normal distribution.

32 answer → Calculator Oxford Exam-style · Q22

[Maximum mark: 5]

A random normal variable X has mean μ and standard deviation 3. Given that $\mathrm{P}(X < 5) = 0.754$

(a) find the value of μ. [3]

(b) Hence find $\mathrm{P}(4 < X < 5)$. [2]

33 answer → Calculator SME Medium · Q8

[Maximum mark: 6]

(a) A machine is used to fill cans of a particular brand of soft drink. The volume, V ml, of soft drink in the cans is normally distributed with mean 330 ml and standard deviation σ ml. Given that 15% of the cans contain more than 333.4 ml of soft drink, find the value of σ. [2]

(b) Find $\mathrm{P}(320 \leq V \leq 340)$. [1]

(c) Six cans of the soft drink are chosen at random. Find the probability that all of the cans contain less than 329 ml of soft drink. [3]

31 ← question *Oxford Ex 14M · Q9*

$\mathrm{P}(X < 3) = 0.65$, $z = 0.3853$; $\sigma = \dfrac{0.1}{0.3853} = \boxed{0.260}\,\text{m}$ **M1 A1 A1**

TI-Nspire CX: Question 31

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**left-tail** area), μ, σ.
For a z-value use $\mu = 0$, $\sigma = 1$. `invNorm(0.65,0,1)`

32 ← question *Oxford Exam-style · Q22*

(a) $z = 0.6871$; $\mu = 5 - 3(0.6871) = \boxed{2.94}$ **M1 A1 A1**

(b) $\boxed{0.116}$ **M1 A1**

TI-Nspire CX: Question 32

`invNorm(0.754,0,1); normCdf(4,5,2.9386,3)`

33 ← question *SME Medium · Q8*

(a) $z = 1.0364$; $\sigma = \dfrac{3.4}{1.0364} = \boxed{3.28}$ **M1 A1**

(b) $\boxed{0.998}$ **A1**

(c) $0.379618^6 = \boxed{0.00302}$ **M1 M1 A1**

TI-Nspire CX: Question 33

`invNorm(0.85,0,1); normCdf(-9E99,329,330,3.2805)^6`

34 answer → Calculator **SME Medium · Q10**

[Maximum mark: 5]

The ages, A in years, at which Liverpool players have made their debuts over the past 20 years are normally distributed with a mean of 22.5 years and a standard deviation of σ years. Given that 10% of Liverpool players make their debuts before turning 20 years old, find: (i) the value of σ, (ii) the probability that a randomly selected player made his debut before his 18th birthday.

35 answer → Calculator **SME Hard · Q2**

[Maximum mark: 7]

(a) The height of the average female giraffe, from a population where the heights follow a normal distribution, is 4.57 m and has a standard deviation of 1.28 m. Find: (i) Q_1 (ii) Q_3 (iii) IQR [4]

(b) The heights, H, of male giraffes are also normally distributed, where $H \sim \mathrm{N}(4.96, \sigma^2)$. If 68% of male giraffes are shorter than the upper quartile of the female giraffe population, find the standard deviation of the male giraffes. [3]

34 ← question *SME Medium · Q10*

(i) $z = -1.2816$; $\sigma = \dfrac{2.5}{1.2816} = \boxed{1.95}$ **M1 A1 A1**

(ii) $\mathrm{P}(A < 18) = \boxed{0.0105}$ **M1 A1**

TI-Nspire CX: Question 34

```
invNorm(0.1,0,1); normCdf(-9E99,18,22.5,1.9508)
```

35 ← question *SME Hard · Q2*

(a) $Q_1 = \boxed{3.71}$, $Q_3 = \boxed{5.43}$, $\mathrm{IQR} = \boxed{1.73}$ **A1 A1 A2**

(b) $z = \mathrm{invNorm}(0.68) = 0.4677$; $\sigma = \dfrac{5.4333 - 4.96}{0.4677} = \boxed{1.01}\,\mathrm{m}$ **M1**

M1 A1

TI-Nspire CX: Question 35

```
invNorm(0.75,4.57,1.28); invNorm(0.68,0,1)
```

36 answer → Calculator **SME Hard · Q5**

[Maximum mark: 9]

(a) The distribution of birth weight of a newborn can be assumed to follow a normal distribution with a mean of 3369 g and a standard deviation of 567 g. A baby is classified as being of low birth weight if its weight is less than 2500 g. Draw a diagram to represent the situation, labelling clearly the mean and the boundary for low birth weight. [2]

(b) Find the expected number of babies from a sample of 1000 that are born with a low birth weight. Give your answer to the nearest integer. [2]

(c) 14% of babies born at a particular hospital weigh more than 4 kg. Show that the babies born at this hospital are not representative of the population. [2]

(d) Given that the standard deviation is the same as the population, find the mean birth weight of a newborn at this particular hospital. Give your answer to the nearest gram. [3]

36 ← question *SME Hard · Q5*

(a) curve centred at 3369, shade left of 2500 **A2**

(b) $1000 \times 0.062684 = 62.7 \approx \boxed{63}$ (integer asked for) **M1 A1**

(c) population: $\mathrm{P}(X > 4000) = 0.133 \neq 0.14$ **M1 R1**

(d) $z = \text{invNorm}(0.86) = 1.0803$; $\mu = 4000 - 567(1.0803) = \boxed{3387}\,\text{g}$

M1 M1 A1

TI-Nspire CX: Question 36

```
normCdf(-9E99,2500,3369,567); invNorm(0.86,0,1)
```

37 answer →

Calculator SME Hard · Q9

[Maximum mark: 7]

(a) From a given population the average person spends on average 122 minutes exercising each week. The data follows a normal distribution and has a standard deviation of σ minutes. It is known that approximately 81.5% of people spend between 50 and 158 minutes exercising each week. Using a sketch of the distribution or otherwise, explain why 36 minutes would provide a good approximation for the value of σ. [4]

(b) From a sample of people within the population, it is known that 15 of them spent less than 65 minutes exercising each week. Using 36 minutes for the standard deviation, find the total number of people within the sample. [3]

38 answer →

Calculator SME Hard · Q11

[Maximum mark: 9]

The length L of carrots in centimetres is normally distributed with mean μ. The table shows probabilities for values of L.

Values of L	$L < 13.9$	$13.9 \leq L \leq 21.7$	$21.7 < L$
$P(L)$	k	0.94	0.03

Any carrots that have a length longer than 20.3 cm are classed as oversized carrots.

(a) Write down the value of k. [2]

(b) Show that the value of μ is 17.8 cm. [2]

(c) Find the probability that when picking a carrot at random, an oversized carrot is chosen. [5]

37 ← question *SME Hard · Q9*

(a) with $\sigma = 36$: $50 = \mu - 2\sigma$, $158 = \mu + \sigma$; area $= 47.5\% + 34\% = 81.5\%$ ✓ **M1 A1 R2**

(b) $\mathrm{P}(X < 65) = 0.056673$; $\dfrac{15}{0.056673} = 264.7 \approx \boxed{265}$ people **M1 M1 A1**

TI-Nspire CX: Question 37

```
normCdf(-9E99,65,122,36)
```

38 ← question *SME Hard · Q11*

(a) $k = 1 - 0.94 - 0.03 = \boxed{0.03}$ **A2**

(b) equal tails, so μ is the midpoint: $\dfrac{13.9 + 21.7}{2} = 17.8$ **M1 A1**

(c) $z = \mathrm{invNorm}(0.97) = 1.8808$; $\sigma = \dfrac{3.9}{1.8808} = 2.0736$ **M1 A1 A1**

$\mathrm{P}(L > 20.3) = \boxed{0.114}$ **M1 A1**

TI-Nspire CX: Question 38

```
invNorm(0.97,0,1); normCdf(20.3,9E99,17.8,2.0736)
```

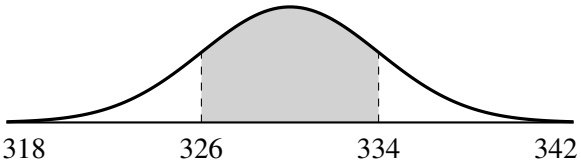
39 answer →

Calculator

SME Very Hard · Q1

[Maximum mark: 7]

The graph shows the normal distribution of the volume, in ml, of drink in a can provided by a manufacturer, with the central 68% of the distribution shaded.



(a) State the mean volume found in a can and the standard deviation. [2]

(b) The probability that a can picked at random has a volume that is greater than v is 0.19. Find v. [2]

(c) A sample of cans of drink were analysed and 27 were found to have a volume of less than 320 ml. Find an estimate for the number of cans of drink that were in the sample. [3]

40 answer →

Calculator

SME Very Hard · Q2

[Maximum mark: 10]

(a) The time in minutes, T, that a customer spends on hold when calling a customer service line follows a normal distribution, where $T \sim \mathrm{N}(\mu, 4^2)$. The probability that a customer spends more than 25 minutes on hold is 0.0228. Find the mean length of time that a customer spends on hold. [3]

(b) Find the interquartile range of the hold times. [2]

(c) Find the time spent on hold that is exceeded by 6% of the population. [2]

(d) From a sample of 200 customers find the probability that exactly 25% of the customers in the sample would spend less than 15 minutes on hold. [3]

39 ← question *SME Very Hard · Q1*

(a) central 68% is $\mu \pm \sigma$: $\boxed{\mu = 330}$ ml, $\boxed{\sigma = 4}$ ml **A1 A1**

(b) left tail 0.81: $v = \boxed{333.5}$ **M1 A1**

(c) $\mathrm{P}(X < 320) = 0.0062097$; $\dfrac{27}{0.0062097} = 4348$ cans **M1 M1 A1**

TI-Nspire CX: Question 39

```
invNorm(0.81,330,4); normCdf(-9E99,320,330,4)
```

40 ← question *SME Very Hard · Q2*

(a) $z = \text{invNorm}(0.9772) = 1.9991$; $\mu = 25 - 4(1.9991) = \boxed{17.0}$ **M1 A1 A1**

(b) $\boxed{5.40}$ **M1 A1**

(c) left tail 0.94: $\boxed{23.2}$ min **M1 A1**

(d) $p = \mathrm{P}(T < 15) = 0.3082$; $X \sim \mathrm{B}(200, p)$: $\mathrm{P}(X = 50) = \boxed{0.0125}$ **M1 M1 A1**

TI-Nspire CX: Question 40

```
invNorm(0.9772,0,1); binomPdf(200,p,50)
```

41 answer → Calculator **SME Very Hard · Q5**

[Maximum mark: 10]

(a) The distance that a honeybee will travel from the hive to collect pollen is normally distributed with an average distance of 1.1 miles. The probability that a honeybee will travel further than 1.6 miles from the hive is 0.00621. Find the variance of the distances that the honeybees will travel from the hive. [4]

(b) 83% of the honeybees travel a distance that is greater than h miles. Find the value of h. [2]

(c) A colony consists of 60 000 honeybees. Calculate the probability that at least 28% but no more than 31% of the honeybees in this colony will stay within 1 mile of the hive. [4]

41 ← question *SME Very Hard · Q5*

(a) $z = \text{invNorm}(0.99379) = 2.4999$; $\sigma = \dfrac{0.5}{2.4999} = 0.2$; variance $\boxed{0.04}$

M1 A1 M1 A1

(b) left tail 0.17: $h = \boxed{0.909}$ **M1 A1**

(c) $p = \text{P}(X < 1) = 0.30854$; $\text{B}(60000, p)$: $\text{P}(16800 \leq Y \leq 18600) =$ $\boxed{0.782}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 41

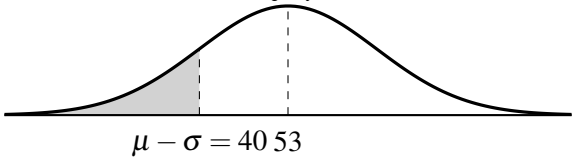
```
binomCdf(60000,0.30854,16800,18600)
```

42 answer →

Calculator **SME Very Hard · Q6**

[Maximum mark: 7]

A teacher sets the same test every year for which the results follow a normal distribution with a mean score of 53 points. Any student who scores lower than one standard deviation below the mean will have to re-sit the test. The diagram shows the boundary $\mu - \sigma$ at a score of 40.



(a) For a class that complete the test, four students are instructed that they will need to re-sit. Find: (i) the standard deviation of the test scores. (ii) an estimate for the total number of students in the class. [3]

(b) A second class of 28 students sit the same test. If they achieve a score of greater than 65, they will achieve a commendation. Find the probability that exactly 3 students in this class achieve a commendation. [4]

43 answer →

Calculator **SME Very Hard · Q7**

[Maximum mark: 9]

(a) The masses, M, of cockroaches are normally distributed with an average mass of 28.8 g. It is known that 0.15% of the population has a mass greater than 51.9 g. Find an approximate value for the variance of the population of cockroaches. [3]

(b) 38 cockroaches are caught in a trap. Using your value for the variance in part (a), find the probability that: (i) exactly 15 of the cockroaches have a mass that is greater than 30 g (ii) more than 26 cockroaches have a mass that is between 26 g and 56 g. [6]

42 ← question *SME Very Hard · Q6*

(a)(i) $\sigma = 53 - 40 = \boxed{13}$ **A1** (ii) $\mathrm{P}(X < \mu - \sigma) = 0.15866$;

$\dfrac{4}{0.15866} = 25.2 \approx \boxed{25}$ students **M1 A1**

(b) $p = \mathrm{P}(X > 65) = 0.17784$; $\mathrm{B}(28, p)$: $\mathrm{P}(Y = 3) = \boxed{0.138}$ **M1 A1**

M1 A1

TI-Nspire CX: Question 42

```
normCdf(65,9E99,53,13); binomPdf(28,0.17784,3)
```

43 ← question *SME Very Hard · Q7*

(a) $z = 2.9677$; $\sigma = \dfrac{23.1}{2.9677} = 7.784$; variance $\approx \boxed{60.6}$ **M1 A1 A1**

(b)(i) $p = \mathrm{P}(M > 30) = 0.4387$; $\mathrm{P}(X = 15) = \boxed{0.113}$ **M1 M1 A1** (ii)

$p = \mathrm{P}(26 < M < 56) = 0.6402$; $\mathrm{P}(Y \geq 27) = \boxed{0.234}$ **M1 M1 A1**

TI-Nspire CX: Question 43

```
invNorm(0.9985,0,1); binomCdf(38,0.64024,27,38)
```

44 answer → Calculator **RV · Q32**

[Maximum mark: 8]

Domino Sugar Bags are packaged by machines. The weights X, in grams, of bags of sugar can be modelled by $X \sim \mathrm{N}(\mu, \sigma^2)$. A bag of sugar is considered to be underweight if it weighs less than 4480 grams.

(a) Given that $\mu = 4536$ and $\sigma = 25$, find the probability that a randomly selected bag of sugar is underweight. [2]

(b) The manufacturer makes the decision that the probability that a bag of sugar is underweight should be 0.001. To do this μ is increased and σ is kept unchanged. Calculate the new value of μ, rounding your answer to the nearest gram. [3]

(c) The manufacturer is unhappy with the way in which this was achieved. The machines are now adjusted to reduce σ and return μ back to 4536 grams. Calculate the new value of σ, rounding your answer to the nearest gram. [3]

45 answer → Calculator **RV · Q33**

[Maximum mark: 7]

Let X and Y be normally distributed with $X \sim \mathrm{N}(135, \sigma^2)$ and $Y \sim \mathrm{N}(105, \sigma^2)$.

(a) Find k such that $\mathrm{P}(Y > k) = \mathrm{P}(X < k)$. [2]

(b) Given that $\mathrm{P}(X > 130) = 0.8944$, find $\mathrm{P}(90 < Y < 110)$. [5]

44 ← question *RV · Q32*

(a) $\boxed{0.0125}$ **M1 A1**

(b) $z = -3.0902$; $\mu = 4480 + 25(3.0902) = \boxed{4557}$ g **M1 A1 A1**

(c) $\sigma = \dfrac{56}{3.0902} = \boxed{18}$ g **M1 A1 A1**

TI-Nspire CX: Question 44

```
invNorm(0.001,0,1)
```

45 ← question *RV · Q33*

(a) same σ, so k is midway: $k = \boxed{120}$ **A2**

(b) $\mathrm{P}(X < 130) = 0.1056$, $z = -1.2502$; $\sigma = \dfrac{5}{1.2502} = 4$ **M1 A1 A1**

$\mathrm{P}(90 < Y < 110) = \boxed{0.894}$ **M1 A1**

TI-Nspire CX: Question 45

```
invNorm(0.1056,0,1); normCdf(90,110,105,4)
```

46 answer → Calculator **RV · Q37**

[Maximum mark: 15]

The time, T minutes, spent travelling to work by employees of a large company is normally distributed with a mean of 46 minutes and a variance of σ^2. The probability that T is less than 30 is 0.118.

(a) Find $\mathrm{P}(30 < T < 46)$. [2]

(b) (i) Find σ, the standard deviation of T. (ii) Hence find the probability that an employee selected at random spends more than 60 minutes travelling to work. [5]

A group of 12 employees is selected at random. Let X represent the number of employees in the group who spend more than 60 minutes travelling to work.

(c) Find $\mathrm{E}[X]$. [3]

(d) Find the probability that exactly four employees in the group spend more than 60 minutes travelling to work. [2]

(e) An employee selected at random spends less than 60 minutes travelling to work. Find the probability that their travelling time is between 30 minutes and 46 minutes. [3]

46 ← question *RV · Q37*

(a) $0.5 - 0.118 = \boxed{0.382}$ **M1 A1**

(b)(i) $z = -1.1850$; $\sigma = \dfrac{16}{1.1850} = \boxed{13.5}$ min **M1 A1 A1** (ii) $\boxed{0.150}$ **M1 A1**

(c) $12 \times 0.150 = \boxed{1.80}$ **M1 M1 A1**

(d) $\mathrm{B}(12, 0.150)$: $\boxed{0.0683}$ **M1 A1**

(e) $\dfrac{0.382}{1 - 0.150} = \boxed{0.449}$ **M1 M1 A1**

TI-Nspire CX: Question 46

```
invNorm(0.118,0,1); normCdf(60,9E99,46,13.5); binomPdf(12,0.15,4)
```

47 answer → Calculator **RV · Q45**

[Maximum mark: 13]

Ali drives from his home to the ferry terminal and then catches the ferry to an island. The time T_1 minutes to drive from his home to the ferry is normally distributed with $T_1 \sim \mathrm{N}(16, 4^2)$. Ali always leaves his house at 07:40. The first ferry that he can catch departs at 08:00.

(a) Find the probability that it will take Ali between 20 to 30 minutes to drive to the ferry. Round your answer to 4 significant figures. [2]

The length of time, T_2 minutes, of the ferry journey is normally distributed with $T_2 \sim \mathrm{N}(25, \sigma^2)$. The probability that the ferry journey takes less than 30 minutes is 0.9520.

(b) Find σ. Round your answer to 1 significant figure. [3]

(c) Find the probability that the ferry journey takes less than 20 minutes. [2]

If Ali misses the first ferry, he can take another one which departs at 08:10. He must arrive at work by 08:30 to be on time. He will not make it to the island if he misses both ferries, and T_1 and T_2 are independent.

(d) Find the probability that Ali will arrive to work on time. Round your answer to 4 significant figures. [4]

(e) This year, Ali will go to work on 214 days. Calculate the number of days Ali is expected to arrive at work on time. [2]

47 ← question *RV · Q45*

(a) $\boxed{0.1584}$ **M1 A1**

(b) $z = 1.6646$; $\sigma = \dfrac{5}{1.6646} = 3.0 \approx \boxed{3}$ **M1 A1 A1**

(c) $\mathrm{P}(T_2 < 20) = \boxed{0.0478}$ **M1 A1**

(d) first ferry ($T_1 < 20$) needs $T_2 < 30$; second ferry ($20 < T_1 < 30$) needs $T_2 < 20$:

$0.84134(0.95221) + 0.15842(0.04779) = \boxed{0.8087}$ **M1 M1 M1 A1**

(e) $214 \times 0.8087 = \boxed{173.1}$ days (not rounded) **M1 A1**

TI-Nspire CX: Question 47

```
normCdf(20,30,16,4); invNorm(0.952,0,1); normCdf(-9E99,20,25,3)
```

48 answer → Calculator **RV · Q35**

[Maximum mark: 18]

The duration, E, in months, of the Everlast brand of car battery, follows a normal distribution with a mean of 42 months and a standard deviation σ months. It is known that 10% of the Everlast batteries will have a larger duration than 60 months.

(a) Find the value of σ. [4]

(b) Find the probability that an Everlast battery lasts less than 12 months. [2]

A battery is considered defective if it lasts less than 12 months. A retailer has purchased a batch of 550 Everlast batteries.

(c) (i) What is the probability the retailer finds at least four defective batteries? (ii) What is the probability the retailer finds at least 8 defective batteries, given that at least four defective batteries have been found? (iii) Find the expected number of defective batteries purchased by the retailer. Round your answer to the nearest integer. [7]

The retailer now tries another brand, Permalast. With D_e the number of defective Everlast batteries in a batch and D_p the number of defective Permalast batteries in a batch, the probability that a single Permalast battery is defective is k, where $0.5 < k < 1$. The retailer buys a new batch of 550 Everlast and 550 Permalast batteries. It is known that $4\text{Var}(D_e) < \text{Var}(D_p) < 5\text{Var}(D_e)$.

(d) Find the range of possible values for k. [5]

48 ← question *RV · Q35*

(a) $z = 1.2816$; $\sigma = \dfrac{18}{1.2816} = \boxed{14.0}$ **M1 A1 M1 A1**

(b) $p = \boxed{0.0163}$ **M1 A1**

(c) $D_e \sim \text{B}(550, 0.016343)$: (i) $\text{P}(D_e \geq 4) = \boxed{0.979}$ (ii) $\dfrac{\text{P}(D_e \geq 8)}{\text{P}(D_e \geq 4)} = \boxed{0.691}$ (iii) $8.99 \approx \boxed{9}$ **M1 A1 M1 M1 A1 M1 A1**

(d) $\text{Var}(D_e) = 8.8417$; $35.37 < 550k(1-k) < 44.21$ with $0.5 < k < 1$:

$\boxed{0.912 < k < 0.931}$ **M1 M1 A1 A1 A1**

TI-Nspire CX: Question 48

`binomCdf(550,0.016343,4,550)`; Graphs: `f1(x)=550x(1-x)` with lines $y = 35.37$ and $y = 44.21$, Intersection.

49 answer →

Calculator **Nov 2012 · P2 · Q6**

[Maximum mark: 8]

In a large city, the time taken to travel to work is normally distributed with mean μ and standard deviation σ. It is found that 4% of the population take less than 5 minutes to get to work, and 70% take less than 25 minutes.

Find the value of μ and of σ.

50 answer →

Calculator **Worksheet · Q2**

[Maximum mark: 7]

The time taken for a student to finish a task is normally distributed with a mean μ and standard deviation σ. It is found that 6% of students take less than 7 minutes to complete the task and 75% take less than 22 minutes.

Find the value of μ and the value of σ.

49 ← question *Nov 2012 · P2 · Q6*

$\mathrm{P}(X < 5) = 0.04,\ \mathrm{P}(X < 25) = 0.70$ **(M1)**

$z_1 = \text{invNorm}(0.04) = -1.75069,\ z_2 = \text{invNorm}(0.70) = 0.524401$ **(A1)(A1)**

$\mu - 1.75069\,\sigma = 5$ and $\mu + 0.524401\,\sigma = 25$ **(M1)(A1)**

$\boxed{\mu = 20.4}$, $\boxed{\sigma = 8.79}$ (minutes) **A1 A1**

TI-Nspire CX: Question 49

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. for each z (`invNorm(area,0,1)`). Simultaneous equations: menu → **3: Algebra** → **2: Solve System of Linear Equations** (2 equations), variables m, s. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

50 ← question *Worksheet · Q2*

$\mathrm{P}(X < 7) = 0.06,\ \mathrm{P}(X < 22) = 0.75$ **(M1)**

$z_1 = \text{invNorm}(0.06) = -1.55477,\ z_2 = \text{invNorm}(0.75) = 0.674490$ **(A1)(A1)**

$\mu - 1.55477\,\sigma = 7$ and $\mu + 0.674490\,\sigma = 22$ **(M1)(A1)**

$\boxed{\mu = 17.5}$, $\boxed{\sigma = 6.73}$ (17.4616…, 6.72868…) **A1 A1**

TI-Nspire CX: Question 50

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. for each z (`invNorm(area,0,1)`). Simultaneous equations: menu → **3: Algebra** → **2: Solve System of Linear Equations** (2 equations), variables m, s. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

51 answer →

Calculator **Worksheet · Q3**

[Maximum mark: 6]

The heights X, in metres, of adult women in a small town can be modelled by a normal distribution. It is given that 8% of women are shorter than 1.55 m and 5% of women are taller than 1.70 m.

Find the mean and standard deviation of the heights of the women.

52 answer →

Calculator **Worksheet · Q4**

[Maximum mark: 6]

It is given that three in ten men take less than 220 minutes to run a marathon. It is also known that one in five men take more than 274 minutes to run the same marathon.

Assume that the time taken, in minutes, for men to run a marathon is modelled by a normal distribution.

Find the mean and standard deviation.

51 ← question *Worksheet · Q3*

Taller than 1.70: 5% is a **right** tail, so use $\mathrm{P}(X < 1.70) = 0.95$.

$\mathrm{P}(X < 1.55) = 0.08,\ \mathrm{P}(X < 1.70) = 0.95$ **(M1)**

$z_1 = \mathrm{invNorm}(0.08) = -1.40507,\ z_2 = \mathrm{invNorm}(0.95) = 1.64485$ **(A1)(A1)**

$\mu - 1.40507\,\sigma = 1.55$ and $\mu + 1.64485\,\sigma = 1.70$ **(M1)(A1)**

$\boxed{\mu = 1.62}$, $\boxed{\sigma = 0.0492}$ m (1.61910, 0.0491815) **A1 A1**

TI-Nspire CX: Question 51

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. for each z (`invNorm(area,0,1)`). Simultaneous equations: menu → **3: Algebra** → **2: Solve System of Linear Equations** (2 equations), variables m, s. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

52 ← question *Worksheet · Q4*

"One in five take more than 274" $\Rightarrow \mathrm{P}(X > 274) = 0.2 \Rightarrow$ left tail 0.8.

$\mathrm{P}(X < 220) = 0.3,\ \mathrm{P}(X < 274) = 0.8$ **(M1)**

$z_1 = \mathrm{invNorm}(0.3) = -0.524401,\ z_2 = \mathrm{invNorm}(0.8) = 0.841621$ **(A1)(A1)**

$\mu - 0.524401\,\sigma = 220$ and $\mu + 0.841621\,\sigma = 274$ **(M1)(A1)**

$\boxed{\mu = 241}$, $\boxed{\sigma = 39.5}$ minutes (240.730, 39.5308) **A1 A1**

TI-Nspire CX: Question 52

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. for each z (`invNorm(area,0,1)`). Simultaneous equations: menu → **3: Algebra** → **2: Solve System of Linear Equations** (2 equations), variables m, s. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

53 answer → Calculator **Worksheet · Q5**

[Maximum mark: 6]

The heights X, in metres, of mature dwarf fruit trees in a tree garden can be modelled by a normal distribution.

It is given that 1% of the dwarf trees are shorter than 2.46 m and 10% are taller than 2.91 m.

Find the mean and standard deviation of the heights of the dwarf trees.

54 answer → Calculator **Oxford Ex 14M · Q3**

[Maximum mark: 6]

$X \sim \mathrm{N}(\mu, \sigma^2)$. Given that $\mathrm{P}(X > 58.39) = 0.0217$ and $\mathrm{P}(X < 41.82) = 0.0287$, find μ and σ.

53 ← question *Worksheet · Q5*

Taller than 2.91: 10% right tail ⇒ use 0.9.

$P(X < 2.46) = 0.01,\ P(X < 2.91) = 0.9$ **(M1)**

$z_1 = \text{invNorm}(0.01) = -2.32635,\ z_2 = \text{invNorm}(0.9) = 1.28155$ **(A1)(A1)**

$\mu - 2.32635\,\sigma = 2.46$ and $\mu + 1.28155\,\sigma = 2.91$ **(M1)(A1)**

$\boxed{\mu = 2.75}$, $\boxed{\sigma = 0.125}$ m (2.75016, 0.124726) **A1 A1**

TI-Nspire CX: Question 53

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. for each z (`invNorm(area,0,1)`). Simultaneous equations: menu → **3: Algebra** → **2: Solve System of Linear Equations** (2 equations), variables m, s. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

54 ← question *Oxford Ex 14M · Q3*

$z_1 = -1.9003,\ z_2 = 2.0198$ **(M1)(A1)(A1)**

$\mu - 1.9003\sigma = 41.82,\ \mu + 2.0198\sigma = 58.39$ **(M1)**

$\boxed{\mu = 49.9}$, $\boxed{\sigma = 4.23}$ **A1 A1**

TI-Nspire CX: Question 54

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

55 answer → Calculator **Oxford Ex 14M · Q4**

[Maximum mark: 6]

A random variable X is normally distributed with mean μ and standard deviation σ, such that $P(X < 89) = 0.90$ and $P(X < 94) = 0.95$. Find μ and σ.

56 answer → Calculator **Oxford Ex 14M · Q10**

[Maximum mark: 8]

It is suspected that the scores in a test are normally distributed. 30% of students scored less than 108 marks on the test, and 20% scored more than 154 marks.

(a) Find the mean and standard deviation of the scores, if they are normally distributed. [6]

(b) 60% of students score more than 117 marks. Explain whether this fact appears to be reasonably consistent with the claim that the scores are normally distributed. [2]

57 answer → Calculator **Oxford Ex 14M · Q11**

[Maximum mark: 6]

Due to variations in manufacturing, the length of wool in a randomly chosen ball of wool can be modelled by a normal distribution. Find the mean and standard deviation in the lengths of wool, given that 95% of balls of wool have lengths exceeding 495 m, and 99% have lengths exceeding 490 m.

55 ← question *Oxford Ex 14M · Q4*

$z_1 = 1.2816,\ z_2 = 1.6449$ **(M1)(A1)(A1)**

$\mu + 1.2816\sigma = 89,\ \mu + 1.6449\sigma = 94$ **(M1)**

$\boxed{\mu = 71.4},\ \boxed{\sigma = 13.8}$ **A1 A1**

TI-Nspire CX: Question 55

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

56 ← question *Oxford Ex 14M · Q10*

(a) $z_1 = -0.5244,\ z_2 = 0.8416$ **(M1)(A1)(A1)**

$\mu - 0.5244\sigma = 108,\ \mu + 0.8416\sigma = 154$ **(M1)**

$\boxed{\mu = 125.7},\ \boxed{\sigma = 33.7}$ **A1 A1**

(b) $\mathrm{P}(X > 117) = 0.601 = 60.1\%$ — consistent **M1 R1**

TI-Nspire CX: Question 56

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

57 ← question *Oxford Ex 14M · Q11*

$\mathrm{P}(X < 495) = 0.05,\ \mathrm{P}(X < 490) = 0.01$

$z_1 = -1.6449,\ z_2 = -2.3263$ **(M1)(A1)(A1)**

$\mu - 1.6449\sigma = 495,\ \mu - 2.3263\sigma = 490$ **(M1)**

$\boxed{\mu = 507.1},\ \boxed{\sigma = 7.34}$ **A1 A1**

TI-Nspire CX: Question 57

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

58 answer → Calculator **Oxford Exam-style · Q24**

[Maximum mark: 14]

A survey is conducted in a large factory. It is found that 27% of the factory workers weigh less than 65 kg and that 25% of the factory workers weigh more than 96 kg. Assume the weights are modelled by a normal distribution with mean μ and standard deviation σ.

(a) (i) Determine two simultaneous linear equations satisfied by μ and σ.
(ii) Find the values of μ and σ. [6]

(b) Find the probability that a factory worker chosen at random weighs more than 100 kg. [2]

The factory has 1000 workers, 630 of which are males. The weights of the males are normally distributed with mean 80.5 and standard deviation 10.1.

(c) If a male factory worker is selected at random, find the probability that he weighs between 75 and 85 kilograms. [2]

(d) Find the expected number of male factory workers that weigh more than 85 kilograms. [2]

(e) The female workers' weights have mean m. Find the value of m. [2]

58 ← question *Oxford Exam-style · Q24*

(a)(i) $\mu - 0.6128\sigma = 65,\ \mu + 0.6745\sigma = 96$ **M1 A1 A1** (ii) $\boxed{\mu = 79.8}$,

$\boxed{\sigma = 24.1}$ **A1 A2**

(b) $\boxed{0.200}$ **M1 A1**

(c) $\boxed{0.379}$ **M1 A1**

(d) $630 \times 0.32796 = \boxed{206.6}$ (not rounded) **M1 A1**

(e) $\dfrac{630(80.5) + 370m}{1000} = 79.757 \Rightarrow \boxed{m = 78.5}$ **M1 A1**

TI-Nspire CX: Question 58

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$. `normCdf(75,85,80.5,10.1)`

59 answer → Calculator **Oxford Exam-style · Q26**

[Maximum mark: 6]

The marks M of a mathematics exam at a large school were normally distributed. Let μ be the mean and σ the standard deviation. Given that 10% of the candidates scored at least 82 marks and 20% scored less than 40, find the value of

(a) the mean, μ [3]

(b) the standard deviation, σ. [3]

60 answer → Calculator **SME Medium · Q9**

[Maximum mark: 6]

(a) The random variable $X \sim \mathrm{N}(\mu, \sigma^2)$. It is known that $\mathrm{P}(X > 36.88) = 0.025$ and $\mathrm{P}(X < 27.16) = 0.1$. Find the values of a and b for which $\mathrm{P}(Z > a) = 0.025$ and $\mathrm{P}(Z < b) = 0.1$, where Z is the standard normal variable. Give your answers correct to 4 decimal places. [2]

(b) Use your answers to part (a), along with the relationship between Z and X, to show that the following simultaneous equations must be true: $\mu + 1.96\sigma = 36.88$ and $\mu - 1.2816\sigma = 27.16$ [2]

(c) By solving the simultaneous equations in (b), determine the values of μ and σ. Give your answers correct to 2 decimal places. [2]

59 ← question *Oxford Exam-style · Q26*

$\mathrm{P}(M < 82) = 0.9$, $\mathrm{P}(M < 40) = 0.2$

$z_1 = 1.2816$, $z_2 = -0.8416$ **(M1)(A1)(A1)**

$\mu + 1.2816\sigma = 82$, $\mu - 0.8416\sigma = 40$ **(M1)**

$\boxed{\mu = 56.6}$, $\boxed{\sigma = 19.8}$ **A1 A1**

TI-Nspire CX: Question 59

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

60 ← question *SME Medium · Q9*

(a) $a = \boxed{1.9600}$, $b = \boxed{-1.2816}$ **A1 A1**

(b) $\dfrac{36.88 - \mu}{\sigma} = 1.96$, $\dfrac{27.16 - \mu}{\sigma} = -1.2816 \Rightarrow$ result **M1 A1**

(c) $\boxed{\mu = 31.00}$, $\boxed{\sigma = 3.00}$ **A1 A1**

TI-Nspire CX: Question 60

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

61 answer → Calculator **SME Hard · Q8**

[Maximum mark: 12]

(a) A company states that the lifespan in hours, H, of the bulbs that they manufacture follows a normal distribution with $H \sim \mathrm{N}(3000, 720^2)$. The company advertises that their lightbulbs exceed a lifespan of h hours. Given that $\mathrm{P}(H < h) = 0.4$, find h. [2]

(b) From a batch of 4000 lightbulbs, calculate the expected number of lightbulbs that will have a lifespan greater than h found in part (a) but less than 3150 hours. [3]

(c) The 5% of light bulbs with the shortest life span are considered to be defective. One of the lightbulbs that is tested has a lifespan of 2213 hours. Determine whether the lightbulb is considered to be defective. Give a reason for your answer. [3]

(d) A second company also manufactures lightbulbs, whose lifespan again follows a normal distribution, $H \sim \mathrm{N}(\mu, \sigma^2)$. Given that $\mathrm{P}(H < 2600) = 0.404$ and $\mathrm{P}(H > 3050) = 0.358$, find the values of μ and σ. [4]

61 ← question *SME Hard · Q8*

(a) $h = \boxed{2818}$ **M1 A1**

(b) $4000 \times 0.18252 = \boxed{730.1}$ **M1 M1 A1**

(c) cut-off $=$ invNorm$(0.05, 3000, 720) = 1816$; $2213 > 1816$: **not defective** **M1 A1 R1**

(d) $z_1 = -0.2430$, $z_2 = 0.3638$: $\boxed{\mu = 2780}$, $\boxed{\sigma = 742}$ **M1 A1 A1 A1**

TI-Nspire CX: Question 61

`invNorm(0.4,3000,720);` Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

62 answer → Calculator **SME Very Hard · Q9**

[Maximum mark: 7]

(a) Within a given population, the length of time that a person can hold their breath is normally distributed. 42% of the population can hold their breath for up to a maximum of 1.2 minutes but only 0.5% of the population can hold their breath for longer than 2.8 minutes. Find (i) the mean, (ii) the variance. [4]

(b) In order to swim through a particular underwater cave without breathing equipment a person must be able to hold their breath for longer than 3.5 minutes. A sample of 1200 people from the population are surveyed. Find the probability that 2 or more people would be able to hold their breath long enough to swim through the cave. [3]

62 ← question *SME Very Hard · Q9*

(a) $z_1 = -0.2019$, $z_2 = 2.5758$: $\mu = \boxed{1.32}$ min, $\sigma = 0.576$, variance $\boxed{0.332}$ **M1 A1 A1 A1**

(b) $p = \mathrm{P}(X > 3.5) = 7.50 \times 10^{-5}$; $\mathrm{B}(1200, p)$: $\mathrm{P}(Y \geq 2) = \boxed{0.00381}$ **M1 M1 A1**

TI-Nspire CX: Question 62

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$. `binomCdf(1200,7.4996E-5,2,1200)`

63 answer → Calculator **RV · Q39**

[Maximum mark: 13]

A survey is conducted in a large technology company. It is found that 25% of the employees drink more than 30 cups of coffee and 4% of the employees drink less than 12 cups of coffee during a five-day week. The number of cups of coffee consumed can be modelled by a normal distribution with mean μ and standard deviation σ.

(a) (i) Determine two linear equations satisfied by μ and σ. (ii) Find the values of μ and σ, giving your answers correct to one decimal place. [6]

(b) Find the probability that an employee drinks more than 40 cups of coffee during a five-day week. [2]

(c) There are 12 employees attending a particular meeting. (i) Find the probability that at least three of the employees drink more than 40 cups of coffee during a five-day week. (ii) Given that at least one of the employees drinks more than 40 cups of coffee, find the probability that there are fewer than three employees drinking more than 40 cups of coffee during a five-day week. [5]

63 ← question *RV · Q39*

(a)(i) $\mu - 1.751\sigma = 12,\ \mu + 0.674\sigma = 30$ **M1 A1 A1** (ii) $\boxed{\mu = 25.0}$, $\boxed{\sigma = 7.4}$ **A1 A2**

(b) $\boxed{0.0213}$ **M1 A1**

(c) $Y \sim \mathrm{B}(12, 0.0213)$: (i) $\boxed{0.00184}$ (ii) $\dfrac{\mathrm{P}(1 \le Y \le 2)}{\mathrm{P}(Y \ge 1)} = \boxed{0.992}$ **M1 A1 M1 M1 A1**

TI-Nspire CX: Question 63

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$. `binomCdf(12,0.0213,3,12)`

64 answer → Calculator **RV · Q41**

[Maximum mark: 12]

The battery lifetime, in hours, of a new model of a smartphone is modelled by a normal distribution with mean μ and standard deviation σ.

(a) Given that 12% of the smartphones have battery lifetime of more than 18 hours and 21% of the smartphones have battery lifetime of less than 13 hours, calculate the value of μ and the value of σ, giving your answers to two significant figures. [6]

(b) A random sample of twenty smartphones is selected. Let X denote the number of smartphones in the sample whose batteries last for more than 18 hours. (i) Calculate $\mathrm{E}[X]$. (ii) Calculate the probability that exactly five of these smartphones have battery lifetime of more than 18 hours. (iii) Determine the most likely value of X. [6]

64 ← question *RV · Q41*

(a) $\mu - 0.806\sigma = 13$, $\mu + 1.175\sigma = 18$: $\boxed{\mu = 15}$ h, $\boxed{\sigma = 2.5}$ h **M1 A1 A1 M1 A1 A1**

(b) $X \sim \mathrm{B}(20, 0.12)$: (i) $\boxed{2.4}$ (ii) $\boxed{0.0567}$ (iii) $\mathrm{P}(X = 2) = 0.274$ is the largest: $\boxed{2}$ **A1 A1 A1 M1 A1 A1**

TI-Nspire CX: Question 64

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$. `binomPdf(20,0.12,5)`

65 answer → Calculator **May 2021 · P2 TZ2 · Q8**

[Maximum mark: 15]

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes.

(a) Given that 2% of the flight times are longer than 82 minutes, find the value of σ. [3]

(b) Find the probability that a randomly selected flight will have a flight time of more than 80 minutes. [2]

(c) Given that a flight between the two cities takes longer than 80 minutes, find the probability that it takes less than 82 minutes. [4]

On a particular day, there are 64 flights scheduled between these two cities.

(d) Find the expected number of flights that will have a flight time of more than 80 minutes. [3]

(e) Find the probability that more than 6 of the flights on this particular day will have a flight time of more than 80 minutes. [3]

65 ← question *May 2021 · P2 TZ2 · Q8*

(a) $P(T > 82) = 0.02 \Rightarrow P(T < 82) = 0.98,\ z = 2.05375$ **(M1)**

$2.05375 = \dfrac{82-75}{\sigma}$ **(A1)** $\boxed{\sigma = 3.41}$ (3.40840…) **A1**

(b) $P(T > 80) = \boxed{0.0712}$ (0.0711930; lower bound is 80) **(M1) A1**

(c) $P(T < 82 \,|\, T > 80) = \dfrac{P(80 < T < 82)}{P(T > 80)} = \dfrac{0.0511930}{0.0711930}$ **(M1)(M1)(A1)** $= \boxed{0.719}$ **A1**

(d) $X \sim B(64, 0.0711930),\ E(X) = 64 \times 0.0711930$ **(M1)(A1)**

$\boxed{4.56}$ flights (**do not round** to 5) **A1**

(e) $P(X > 6) = P(X \geq 7) = 1 - P(X \leq 6) = 1 - 0.830880$ **(M1)(A1)** $= \boxed{0.169}$ **A1**

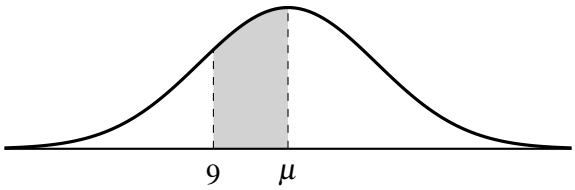
TI-Nspire CX: Question 65

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. (a) `invNorm(0.98,0,1)`. menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) (b) `normCdf(80,∞,75,3.4084)`. (e) menu → **5: Probability** → **5: Distributions** → **E: Binomial Cdf**: $n = 64$, $p = 0.071193$, lower 7, upper 64 → 0.169.

66 answer → Calculator **May 2017 · P2 TZ1 · Q9**

[Maximum mark: 15]

A random variable X is normally distributed with mean, μ. In the following diagram, the shaded region between 9 and μ represents 30% of the distribution.



(a) Find $P(X < 9)$. [2]

The standard deviation of X is 2.1.

(b) Find the value of μ. [3]

The random variable Y is normally distributed with mean λ and standard deviation 3.5. The events $X > 9$ and $Y > 9$ are independent, and $P((X > 9) \cap (Y > 9)) = 0.4$.

(c) Find λ. [5]

(d) Given that $Y > 9$, find $P(Y < 13)$. [5]

66 ← question *May 2017 · P2 TZ1 · Q9*

(a) $P(X < \mu) = 0.5$, so $P(X < 9) = 0.5 - 0.3 = \boxed{0.2}$ **(M1) A1**

(b) $z = \text{invNorm}(0.2) = -0.841621$ **(A1)**

$-0.841621 = \dfrac{9-\mu}{2.1}$ **(M1)** $\boxed{\mu = 10.8}$ (10.7674) **A1**

(c) $P(X > 9) = 0.8$ **(A1)**; independent: $0.8 \times P(Y > 9) = 0.4$ **(M1)(A1)**

$P(Y > 9) = 0.5$ **A1**, so 9 is the mean: $\boxed{\lambda = 9}$ **A1**

(d) $P(9 < Y < 13) = 0.373450$ **(A2)**

$P(Y < 13 \,|\, Y > 9) = \dfrac{0.373450}{0.5}$ **(M1)(A1)** $= \boxed{0.747}$ **A1**

TI-Nspire CX: Question 66

menu → **5: Probability** → **5: Distributions** → **3: Inverse Normal**: Area (**always the left-tail area**), μ, σ. For a z-value use $\mu = 0$, $\sigma = 1$. (b) `invNorm(0.2,0,1)` $= -0.841621$. menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) (d) `normCdf(9,13,9,3.5)` $= 0.373450$.

67 answer →

Calculator

Nov 2017 · P2 · Q8

[Maximum mark: 14]

Adam is a beekeeper who collected data about monthly honey production in his bee hives. The data for six of his hives is shown in the following table.

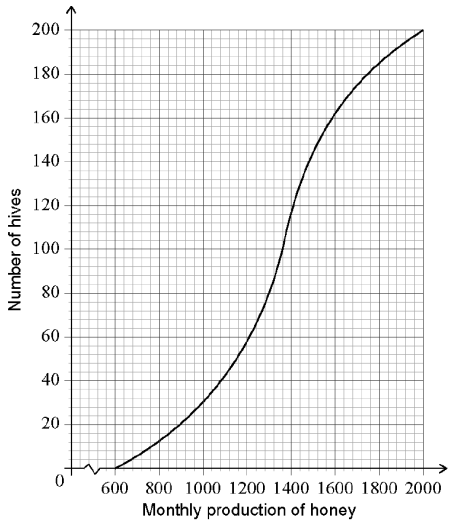
Number of bees (N)	190	220	250	285	305	320
Monthly honey production in grams (P)	900	1100	1200	1500	1700	1800

The relationship between the variables is modelled by the regression line with equation $P = aN + b$.

(a) Write down the value of a and of b. [3]

(b) Use this regression line to estimate the monthly honey production from a hive that has 270 bees. [2]

Adam has 200 hives in total. He collects data on the monthly honey production of all the hives. This data is shown in the following cumulative frequency graph.



Adam's hives are labelled as low, regular or high production, as defined in the following table.

Type of hive	low	regular	high
Monthly honey production in grams (P)	$P \leq 1080$	$1080 < P \leq k$	$P > k$

(c) Write down the number of low production hives. [1]

Adam knows that 128 of his hives have a regular production.

(d) Find (i) the value of k; (ii) the number of hives that have a high production. [5]

(e) Adam decides to increase the number of bees in each low production hive. Research suggests that there is a probability of 0.75 that a low production hive becomes a regular production hive. Calculate the probability that 30 low production hives become regular production hives. [3]

67 ← question *Nov 2017 · P2 · Q8*

(a) $\boxed{a = 6.96}$, $\boxed{b = -455}$ (6.96104, −454.805) **(M1) A1 A1**

(b) $P = 6.96104(270) - 454.805 = 1424.67 \approx \boxed{1420}$ g **(M1) A1**

(c) read at 1080: $\boxed{40}$ hives **A1**

(d)(i) $40 + 128 = 168$ hives have $P \leq k$ **(M1)(A1)**; read across at 168: $\boxed{k = 1640}$ **A1**

(ii) $200 - 168 = \boxed{32}$ hives **(M1) A1**

(e) $X \sim \mathrm{B}(40, 0.75)$, $\mathrm{P}(X = 30) = \binom{40}{30} 0.75^{30} 0.25^{10}$ **(M1)(A1)** $= \boxed{0.144}$ **A1**

TI-Nspire CX: Question 67

(a) home → **Lists & Spreadsheet**: N in A, P in B. menu → **4: Statistics** → **1: Stat Calculations** → **3: Linear Regression (mx+b)**: X List a[], Y List b[] → $m = 6.96104$, $b = -454.805$. (e) menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: $n = 40$, $p = 0.75$, X value 30 → 0.144364.

68 answer → Calculator **Practice**

[Maximum mark: 7]

The masses of apples from an orchard are normally distributed with mean 85 g and standard deviation 4 g.

(a) Find the probability that an apple has a mass of more than 80 g. [2]

(b) Find the probability that an apple has a mass between 80 g and 86 g. [2]

(c) A box contains 250 apples. Find the expected number of apples in the box with a mass of more than 86 g. [3]

69 answer → Calculator **Practice**

[Maximum mark: 5]

The scores in a test taken by 120 students are normally distributed with mean 72 and standard deviation 5.

(a) Find the expected number of students who score between 65 and 80. [3]

(b) Two students are chosen at random. Find the probability that both score more than 80. [2]

68 ← question *Practice*

(a) Continuous, so the bound is exactly 80 (not 81): $P(X > 80) = \boxed{0.894}$ **M1 A1**

(b) $P(80 < X < 86) = \boxed{0.493}$ (same as $P(80 \leq X \leq 86)$) **M1 A1**

(c) $P(X > 86) = 0.401294$ **A1**; $E = 250 \times 0.401294$ **M1** $= \boxed{100.3}$ apples (do not round to 100) **A1**

TI-Nspire CX: Question 68

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) (a) `normCdf(80,9E99,85,4)`; (b) `normCdf(80,86,85,4)`; (c) `250×normCdf(86,9E99,85,4)`.

69 ← question *Practice*

(a) $P(65 < X < 80) = 0.864444$ **M1 A1**; $120 \times 0.864444 = \boxed{103.7}$ students (103.733, not rounded) **A1**

(b) $P(X > 80) = 0.0547993$; $0.0547993^2 = \boxed{0.00300}$ **M1 A1**

TI-Nspire CX: Question 69

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower Bound, Upper Bound, μ, σ. (Use -9×10^{99} or $-$ E 99 for $-\infty$.) `normCdf(65,80,72,5)` $= 0.864444$; `normCdf(80,9E99,72,5)` $= 0.0547993$.

70 answer → Calculator SME Very Hard · Q3

[Maximum mark: 11]

(a) The stem heights, H in cm, of a particular variety of tulip follow a distribution where $H \sim \mathrm{N}(60.1, 57.76)$. The probability that a randomly selected tulip will have a stem height that is greater than h is 0.72. Find h. [3]

(b) A tulip is selected at random. It is known that the height of the stem is more than 62 cm. Find the probability that the stem height is taller than 64 cm. [4]

(c) Leila buys a bunch of 24 tulips. Calculate the probability that at least 10 of them have a stem height of less than 57 cm. [4]

71 answer → Calculator RV · Q38

[Maximum mark: 15]

The heights X, in metres, of trees in a forest are normally distributed with a mean of 56.7 m and a standard deviation of 18.2 m.

(a) A tree is selected at random. Find the probability that the tree is: (i) between 65 m and 85 m; (ii) at least 70 m. [3]

(b) The tallest 10% of trees are considered giant trees and the smallest 15% of trees are considered small trees. (i) Calculate the minimum height for a tree to be considered giant. (ii) Calculate the maximum height for a tree to be considered small. [4]

(c) Four trees are selected at random. Find the probability that two trees are giant and the other two are small. [4]

(d) Twenty trees are selected at random. Find the probability that: (i) no trees are giant; (ii) at least three trees are giant. [4]

70 ← question *SME Very Hard · Q3*

$\sigma = \sqrt{57.76} = 7.6$

(a) left tail 0.28: $h = \boxed{55.7}$ **M1 A1 A1**

(b) $\dfrac{\mathrm{P}(H > 64)}{\mathrm{P}(H > 62)} = \dfrac{0.30392}{0.40129} = \boxed{0.757}$ **M1 M1 A2**

(c) $p = 0.34168$; $\mathrm{B}(24, p)$: $\mathrm{P}(X \geq 10) = \boxed{0.283}$ **M1 M1 M1 A1**

TI-Nspire CX: Question 70

```
invNorm(0.28,60.1,7.6); binomCdf(24,0.34168,10,24)
```

71 ← question *RV · Q38*

(a)(i) $\boxed{0.264}$ (ii) $\boxed{0.232}$ **A1 M1 A1**

(b)(i) $\text{invNorm}(0.9, 56.7, 18.2) = \boxed{80.0}$ m (ii) $\boxed{37.8}$ m **M1 A1 M1 A1**

(c) $\binom{4}{2} \times 0.1^2 \times 0.15^2 = 6(0.01)(0.0225) = \boxed{0.00135}$ **M1 M1 M1 A1**

(d) $Y \sim \mathrm{B}(20, 0.1)$: (i) $\boxed{0.122}$ (ii) $\boxed{0.323}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 71

```
invNorm(0.9,56.7,18.2); binomCdf(20,0.1,3,20)
```

72 answer → Calculator **RV · Q44**

[Maximum mark: 16]

The mass M, in grams, of oranges is normally distributed with mean μ. The table below shows probabilities for values of M.

Values of M	$M < 110$	$110 \leq M \leq 220$	$M > 220$
$P(M)$	0.02	0.96	k

(a) (i) Write down the value of k. (ii) Show that $\mu = 165$. [3]

(b) Find $P(M < 120)$. [5]

(c) The oranges are packed in boxes of fifteen. Any orange with a mass less than 120 grams is classified as small. Find the probability that a box of oranges selected at random contains at most two small oranges. [3]

(d) A wooden crate contains 20 boxes of oranges. A crate is selected at random. (i) Find the expected number of boxes in this crate that contain at most two small oranges. (ii) Find the probability that at least 18 boxes in this crate contain at most two small oranges. [5]

72 ← question *RV · Q44*

(a)(i) $\boxed{k = 0.02}$ (ii) equal tails: $\mu = \dfrac{110+220}{2} = 165$ **A1 M1 A1**

(b) $z = -2.0537$; $\sigma = \dfrac{55}{2.0537} = 26.78$; $\mathrm{P}(M < 120) = \boxed{0.0464}$ **M1 A1**

M1 A1 A1

(c) $\mathrm{B}(15, 0.0464)$: $\mathrm{P}(X \leq 2) = \boxed{0.970}$ **M1 A2**

(d) $\mathrm{B}(20, 0.970)$: (i) $\boxed{19.4}$ boxes (ii) $\mathrm{P}(Y \geq 18) = \boxed{0.979}$ **M1 A1 M1**

M1 A1

TI-Nspire CX: Question 72

```
invNorm(0.02,0,1); binomCdf(15,0.0464,0,2); binomCdf(20,0.97,18,20)
```

73 answer → Calculator **RV · Q48**

[Maximum mark: 12]

Farmer Hitoshi grows Kamo eggplants and the weights of his eggplants are normally distributed with a mean of 284 g and standard deviation of 66 g.

(a) (i) Calculate the percentage of Hitoshi's eggplants that weigh between 230 g and 320 g. (ii) Hitoshi has 150 eggplants to take to market. Find the expected number of eggplants weighing more than 320 g. Give your answer correct to the nearest integer. (iii) Find the probability that at least 45 of the 150 eggplants weigh more than 320 g. [6]

Farmer Antonio grows Barbarella eggplants and the weights of his eggplants are normally distributed with a mean of 402 g. Antonio only takes to market eggplants that weigh more than 320 g. Over a period of time, Antonio finds that he has to reject 1 in 12 eggplants due to them being underweight.

(b) (i) Find the standard deviation of the weights of Antonio's eggplants. (ii) Antonio has 250 eggplants to take to the market. Find the expected number of eggplants weighing more than 410 g. Give your answer correct to the nearest integer. [6]

73 ← question *RV · Q48*

(a)(i) $0.501 \Rightarrow \boxed{50.1\%}$ (ii) $150 \times 0.29272 = 43.9 \approx \boxed{44}$ (iii)

B$(150, 0.29272)$: $\mathrm{P}(X \geq 45) = \boxed{0.453}$ **A2 M1 A1 M1 A1**

(b)(i) $z = \text{invNorm}(\frac{1}{12}) = -1.3830$; $\sigma = \frac{82}{1.3830} = \boxed{59.3}$ g **M1 A1 A1**

(ii) $250 \times 0.44634 = 111.6 \approx \boxed{112}$ **M1 M1 A1**

TI-Nspire CX: Question 73

```
normCdf(230,320,284,66); binomCdf(150,0.29272,45,150); invNorm(1/12,0,1)
```

74 answer → Calculator **RV · Q50**

[Maximum mark: 15]

A health study on body fat percentage is conducted on adult male participants in two dietary groups. The participants in Group A consume less than 52 grams of sugar daily. Their body fat percentages are normally distributed with mean 15% and standard deviation 3%.

(a) A person is selected at random from Group A. Find the probability that their body fat percentage is greater than 20%. [2]

The participants in Group B consume more than 52 grams of sugar daily. Their body fat percentages are normally distributed with mean 24% and standard deviation σ%.

(b) The probability that the body fat percentage of a person in Group B is less than 20% is 0.0548. Find the value of σ. [4]

In the study, there are 160 participants in Group A and 80 participants in Group B.

(c) A randomly selected participant has a body fat percentage greater than 20%. Find the probability that the participant is in Group A. [6]

(d) Twelve of the participants with body fat percentages greater than 20% are selected at random. Find the probability that at least two of them are in Group A. [3]

74 ← question *RV · Q50*

(a) $\boxed{0.0478}$ **M1 A1**

(b) $z = -1.6000$; $\sigma = \dfrac{4}{1.6} = \boxed{2.5}$ **M1 A1 M1 A1**

(c) $\mathrm{P}(B, > 20) = 0.9452$: $\dfrac{160(0.04779)}{160(0.04779)+80(0.9452)} = \boxed{0.0918}$ **M1**

M1 M1 A1 A2

(d) $\mathrm{B}(12, 0.0918)$: $\mathrm{P}(X \geq 2) = \boxed{0.303}$ **M1 M1 A1**

TI-Nspire CX: Question 74

```
invNorm(0.0548,0,1); binomCdf(12,0.091836,2,12)
```

75 answer → Calculator **RV · Q52**

[Maximum mark: 15]

A pencil manufacturer measures the length, L mm, of their new batch of pencils. The lengths of the pencils are normally distributed, where $L \sim \mathrm{N}(110, \sigma^2)$.

(a) Find $\mathrm{P}(110 - 2\sigma < L < 110 + \sigma)$. [3]

The probability that the length of a pencil is less than 116 mm is 0.95.

(b) Find the value of σ. [3]

All pencils with length at least k mm are classified as large pencils.

(c) A pencil is chosen at random. The probability it is a large pencil is 0.6. Find the value of k. [3]

(d) (i) A pencil is chosen at random from the large pencils. Find the probability that the length of this pencil is less than 111 mm. (ii) Twenty pencils are chosen at random from the large pencils. Find the probability that at least three pencils have a length less than 111 mm. [6]

75 ← question *RV · Q52*

(a) $\mathrm{P}(-2 < Z < 1) = \boxed{0.819}$ **M1 A2**

(b) $z = 1.6449$; $\sigma = \dfrac{6}{1.6449} = \boxed{3.65}$ **M1 A1 A1**

(c) left tail 0.4: $k = \boxed{109.1}$ **M1 A1 A1**

(d)(i) $\dfrac{\mathrm{P}(109.08 < L < 111)}{0.6} = \boxed{0.347}$ (ii) $\mathrm{B}(20, 0.34669)$: $\mathrm{P}(X \geq 3) = \boxed{0.987}$ **M1 M1 A1 M1 M1 A1**

TI-Nspire CX: Question 75

```
normCdf(-2,1,0,1); invNorm(0.4,110,3.6477); binomCdf(20,0.34669,3,20)
```

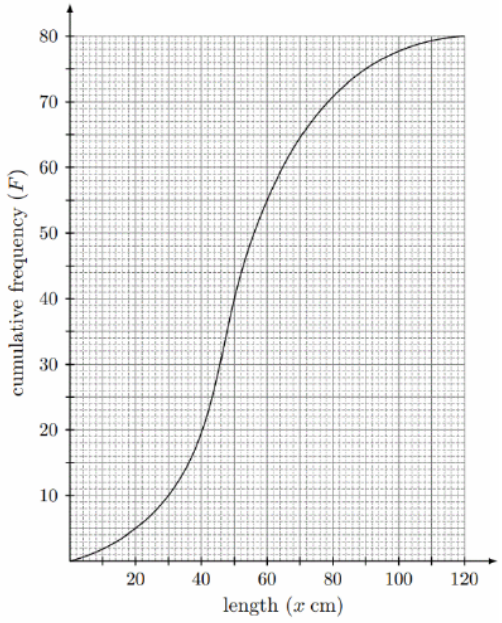
76 answer →

Calculator

School HW Extension · Q1

[Maximum mark: 15]

The following cumulative frequency diagram shows the lengths of 80 flower stems in a garden, in centimetres.



(a) (i) Write down the median length. (ii) Find the percentage of flower stems that are 60 cm or greater. [4]

The same data is presented in the following frequency table.

Length (x cm)	$0 \leq x \leq 30$	$30 \leq x \leq 60$	$60 \leq x \leq 90$	$90 \leq x \leq 120$
Frequency	10	p	20	q

(b) Find the value of: (i) p; (ii) q. [4]

(c) Use the values from the frequency table to estimate the mean and standard deviation of the lengths. [3]

A flower with stem length of 60 cm or greater is considered as a mature flower.

(d) Given that a randomly selected flower is mature, find the probability that its stem length is 90 cm or greater. [4]

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(a)(i) $\boxed{50}$ cm (ii) $\dfrac{80-55}{80} = 0.3125 \Rightarrow \boxed{31.25\%}$ **A1 M1 A2**

(b) (i) $10+p=55 \Rightarrow \boxed{p=45}$ (ii) $75+q=80 \Rightarrow \boxed{q=5}$ **M1 A1 M1**

A1

(c) mid-values 15, 45, 75, 105: $\bar{x} = \boxed{52.5}$ cm, $\sigma = \boxed{22.5}$ cm **M1 A1 A1**

(d) $\dfrac{5/80}{25/80} = \boxed{0.2}$ **M1 M1 M1 A1**

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Lists & Spreadsheet: mid-values in A, frequencies in B; menu 4: Statistics → 1: Stat Calculations → 1: One-Variable Statistics (X1 a[], Frequency b[]).